

ECE1387: Linear System Solver Information (Fall 2025)

Download SuiteSparse from this link. I tested version 5.10.1 on remote.ecf.utoronto.ca:
[Link](#)

We'll use the UMFPACK package in SuiteSparse as the solver for Assignment #2.

Follow this recipe carefully to compile the solver and see a small example of its use... I verified this indeed works on ECF.

On ECF, when you unarchive SuiteSparse, go to the directory: SuiteSparse/SuiteSparse_config and edit the file, SuiteSparse_config.mk. Find the line that begins UMFPACK_CONFIG (line 289), uncomment it, and ensure that -DNBLAS -DNCHOLMOD are added to the end of this line. BLAS is a library to accelerate math, but it is not installed on ECF and we do not need to focus on run-time for Assignment #2, so we don't need it.

Then, in that same directory (SuiteSparse_config), type:

```
make library
```

Then, go up a directory and go into the AMD sub-directory, type:

```
make library
```

Then, go up a directory and go into the UMFPACK sub-directory, type:

```
cd Lib
```

Edit the Makefile and find the line that begins LDLIBS +=

Remove \$(BLAS) from the end of the line. Exit your editor.

Return to the UMFPACK directory (up one level) and type:

```
make library
```

You should have successfully compiled UMFPACK...

Finally, to compile and test an example that solves a linear system with UMFPACK, enter the Demo sub-directory of the UMFPACK directory. In there, take a look at umfpack_simple.c which formulates and solves a small linear system.

Edit the Makefile in the Demo sub-directory and remove \$(BLAS) from the line that begins "LIBS =". Then type:

```
make umfpack_simple
```

Then you can run:

```
./umfpack_simple
```

To understand the UMFPACK matrix format and solving routines, the only documentation you need to look at is Section 2 of this document:

SuiteSparse/UMFPACK/Doc/UMFPACK_QuickStart.pdf